

Intro to Deep Learning (Fall 2025) – Final Project

The final project is designed to emphasize your **practical understanding** of the material learned throughout the course.

Each group must consist of **2–3 students** (exceptions may be approved in special cases).

We offer **five predefined project options**, as well as an additional option to **propose your own idea** (subject to my approval).

On **Wednesday at 18:30**, a project registration form will open on Moodle.

Each project has a **limited number of slots**, and registration will be on a **first-come, first-served** basis.

Submission Requirements

Each group must submit:

- A **final report**
- A **live presentation** (attendance is mandatory) during the last 1–2 weeks of the semester
- A **GitHub repository** containing all your code
- A **README file** with installation instructions and a **demo script** for running (inference) your model on **train, validation and new test images**.

These are **real, practical projects** built on **real data**, giving you **hands-on experience** that **can directly strengthen your portfolio** with a meaningful additional project.

Grading Breakdown

Grading is based on a weighted combination of several criteria.

While the weighting is fixed, the score in each category includes a degree of subjective evaluation.

Criterion	Weight
Originality & creativity of your solution, integration of new ideas, and execution quality	20%
Effort and amount of work – 20%	20%
Final report quality (Abstract, Introduction, Related Work, Method, Experiments, Ablation Study, Limitations)	20%
Reproducibility & accessibility (Installation, GitHub code, optional webpage/app demo)	10%
Presentation quality on the project presentation days	15%
Performance on my test data , compared to other groups	15%

The list of the projects:

1. Project 1: Chessboard square Classification and Board-State Reconstruction (Real Data)
2. Project 2: Chess Synthetic-to-Real Generalization for Chessboard Classification.
3. Project 3: Chess Synthetic-to-Real Image Translation for Chessboard real-data generation.
4. Project 4: Campus Image-to-GPS Regression for localization and navigation tasks.
5. Project 5: CUDA based Wavelet Convolution Implementation.
6. Open Project (subject to my approval).

Project 3: Synthetic-to-Real Image Translation for Chessboard Rendering

Description:

In this project, you will train a **chessboard image generation model** whose goal is to transform synthetic chessboard renders into images that visually match real-world appearance while **preserving** the geometry and board layout (piece positions, camera pose, etc.).

This is an **image-to-image translation** task focused on closing the **visual** domain gap between synthetic and real images.

Unlike **Project 2**, which trains a classifier on synthetic images and evaluates its ability to generalize to real images, **Project 3 is not about classification at all**.

Instead, your task here is to **generate** realistic images from synthetic inputs, improving the synthetic data quality itself.

The output is an *image*, not a set of labels.

You will receive chessboard generation code (that I created) using Blender and PyBlender, along with a predefined set of 3D chess pieces and a chessboard. If you need additional synthetic variation (e.g., different piece designs), you may adapt the code and download more Blender chess sets (a website will be provided).

Key Components:

- Using the provided PyBlender generation script to create your synthetic dataset
- Converting synthetic chessboard renders into **realistic-looking** images.
- Preserving geometric structure: piece identity, board layout, etc.
- Exploring image-to-image translation methods (GANs, CycleGAN, diffusion-based approaches, etc.)
- Evaluating how well the translated images match the real domain visually (qualitatively and/or quantitatively)

